

K8S exercises

Get connected to the Cluster



```
az account set --subscription a476a434-0f2f-45db-bf17-cf27

az aks get-credentials --resource-group rg-containerworksh

kubectl get ns
```

Create your own namespace



```
kubectl create namespace <name>

kubectl config set-context --current --namespace=<name>
```

Create and explore a pod



```
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
spec:
  containers:
    - name: nginx
      image: nginx
```

```
ports:
  - containerPort: 80
```

▼

```
kubectl apply -f pod.yml
kubectl get pods
kubectl describe pods nginx-pod
kubectl logs nginx-pod
kubectl delete pod nginx-pod
```

Create a Deployment

▼

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deploy
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx
          ports:
            - containerPort: 80
```

Apply deployment

▼
kubectl apply -f deployment.yaml
kubectl get deployments
kubectl get rs
kubectl get pods <-l app=nginx>

Scale the deployment

▼
kubectl scale deployment nginx-deploy --replicas=5

Update deployment

▼

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deploy
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:1.25
          ports:
            - containerPort: 80
```



```
kubectl get pods -w  
kubectl rollout history deployment nginx-deploy  
kubectl rollout history deployment nginx-deploy --revision
```

Break the deployment



```
apiVersion: apps/v1  
kind: Deployment  
metadata:  
  name: nginx-deploy  
spec:  
  replicas: 2  
  selector:  
    matchLabels:  
      app: nginx  
  template:  
    metadata:  
      labels:  
        app: nginx  
    spec:  
      containers:  
        - name: nginx  
          image: nginx:badtag  
          ports:  
            - containerPort: 80
```

Repair the deployment



```
kubectl set image deployment/nginx-deploy nginx=nginx
```

Expose deployment with a service

Deploy the service

```
▼  
  
apiVersion: v1  
kind: Service  
metadata:  
  name: nginx-service  
spec:  
  selector:  
    app: nginx  
  ports:  
    - protocol: TCP  
      port: 80  
      targetPort: 80  
  type: ClusterIP
```

```
▼  
  
kubectl apply -f service.yml  
kubectl get svc
```

Test the service with ClusterIP

```
▼  
  
kubectl run test-client --image=busybox --restart=Never -i  
wget -q0- nginx-service  
exit
```

Probes

Setting up correct probes



```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deploy
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx
          ports:
            - containerPort: 80
          livenessProbe:
            httpGet:
              path: /
              port: 80
            initialDelaySeconds: 5
            periodSeconds: 10
          readinessProbe:
            httpGet:
              path: /
              port: 80
            initialDelaySeconds: 5
            periodSeconds: 5
```



```
kubectl describe pod nginx-deploy
```

Breaking the probes

```
livenessProbe:
  httpGet:
    path: /wrong-path
    port: 80
  initialDelaySeconds: 5
  periodSeconds: 10
```

=> keeps on restarting the pod

```
kubectl rollout undo deployment/nginx-deploy
```

```
readinessProbe:
  httpGet:
    path: /wrong-path
    port: 80
  initialDelaySeconds: 5
  periodSeconds: 5
```

=> traffic will never be routed to this pod, pod is never ready, no restarts

```
kubectl run -it busybox --image=busybox --restart=Never --
# Inside busybox
wget --timeout=1 nginx-service
```

Loadbalancer service



```
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  type: LoadBalancer
  selector:
    app: nginx
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
```



```
kubectl apply -f ./loadbalancer-service.yml
kubectl get svc
```

Edit HTML in pod



```
kubectl get pods

kubectl exec --stdin --tty <pod-name> -- /bin/sh

cd /usr/share/nginx/html
printf 'I was here' > index.html.html
```

Deploy with ConfigMap

▼

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: custom-html
data:
  index.html: |
    <!DOCTYPE html>
    <html lang="en">
    <head>
      <meta charset="UTF-8">
      <meta name="viewport" content="width=device-width, i
      <title>Welcome to Kubernetes</title>
      <style>
        body {
          background: #f9fafb;
          font-family: 'Segoe UI', Tahoma, Geneva, Verdana
          margin: 0;
          padding: 0;
          display: flex;
          align-items: center;
          justify-content: center;
          height: 100vh;
        }
        .card {
          background: white;
          padding: 2rem 3rem;
          box-shadow: 0 4px 12px rgba(0, 0, 0, 0.1);
          border-radius: 12px;
          text-align: center;
          max-width: 500px;
        }
        h1 {
          color: #2563eb;
          font-size: 2rem;
          margin-bottom: 1rem;
        }
      </style>
    </head>
    <body>
      <div class="card">
        <h1>Welcome to Kubernetes</h1>
      </div>
    </body>
  </html>
```

```
    p {
      color: #4b5563;
      font-size: 1.1rem;
    }
    .footer {
      margin-top: 2rem;
      font-size: 0.9rem;
      color: #9ca3af;
    }
  </style>
</head>
<body>
  <div class="card">
    <h1>👋 Hello from Kubernetes!</h1>
    <p>This web page is being served from a ConfigMap.
    <div class="footer">
      Author: Baptiste
      Served by NGINX on AKS
    </div>
  </div>
</body>
</html>
```



```
kubectl apply -f .\configmap.yml
```



```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deploy
spec:
  replicas: 1
```

```
selector:
  matchLabels:
    app: nginx
template:
  metadata:
    labels:
      app: nginx
  spec:
    containers:
      - name: nginx
        image: nginx
        ports:
          - containerPort: 80
        volumeMounts:
          - name: html
            mountPath: /usr/share/nginx/html/index.html
            subPath: index.html
    volumes:
      - name: html
        configMap:
          name: custom-html
```



```
kubectl apply -f .\deployment-configmap.yml
```